Google Cloud H2S

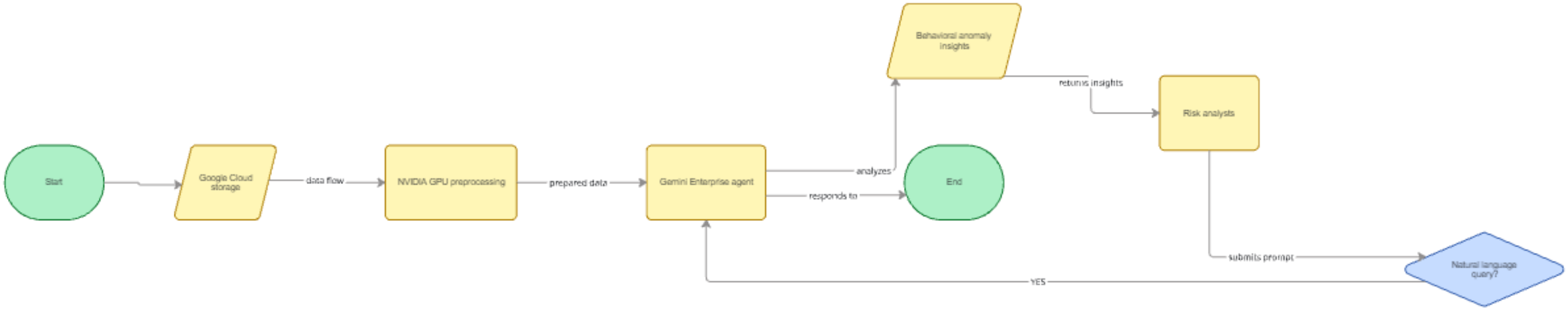
Gen AI Academy
APAC Edition

Build in APAC. Build for the world.

Participant Details

a. **Participant Name:** Binish Abbi
b. **Problem Statement:** Corporate audit teams face massive delays and processing bottlenecks trying to ingest, clean, and profile multi-million line transaction ledgers using legacy, CPU-bound infrastructures. This latency leaves enterprises deeply exposed to missed fraud, financial variances, and compliance reporting gaps.

FinPulse AI is an enterprise financial decision intelligence solution. It combines Google Cloud data storage with NVIDIA GPU-accelerated preprocessing layers to eliminate dataframe processing delays. It interfaces directly with a conversational Gemini Enterprise agent, enabling non-technical risk analysts to uncover deep behavioral anomalies via natural language prompts.



Solution Breakdown:

- **Approach to Problem Statement:** The architecture establishes a repeatable data pipeline that ingests raw accounting ledgers into Google Cloud infrastructure. To meet the core acceleration rules, it applies an NVIDIA RAPIDS layer to clean and parse huge transaction matrices at the hardware layer before serving structured outputs to a language interface.
- **Real-World Problem & Practical Impact:** It directly solves data backlogs within commercial finance departments. The platform drops data-to-insight timelines from several weeks to seconds, ensuring fast regulatory responses and lowering financial loss exposure.
- **Core Architecture Workflow:** Raw Ledger Ingestion $\rightarrow$ Google Cloud Storage $\rightarrow$ NVIDIA RAPIDS Preprocessing $\rightarrow$ Google BigQuery Warehousing $\rightarrow$ Gemini Enterprise Agent

Opportunities & USP

- **Differentiation from Existing Ideas:** Traditional financial audit tools run exclusively on rigid, hardcoded math rules that only flag simple duplicates or typos. FinPulse AI couples high-speed matrix computation with conversational semantic reasoning to identify complex, multi-account behavioral anomalies that legacy applications miss.
- **USP of the Proposed Solution:** Unifies sub-second data-frame cleaning via NVIDIA cuDF with contextual ledger querying through Gemini, bringing zero-code analytics to enterprise auditing.

List of Features Offered by the Solution

- **GPU-Accelerated Matrix Cleaning:** Preprocesses and handles multi-gigabyte transactional ledgers instantly without system or browser performance crashes.
- **Natural-Language-to-SQL Conversion:** Automatically converts conversational text questions directly into working database scripts behind the scenes.
- **Semantic Anomaly Profiling:** Uses context-aware reasoning to spot hidden behavioral trends, transaction timing clusters, and collusion signs across disparate accounts.
- **Reproducible Audit Trails:** Generates permanent, transparent query logs to support corporate compliance validation.

Process flow diagram or Use-case diagram

[Auditor inputs query in natural language]

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▼

[Gemini translates question to SQL script]

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▼

[NVIDIA cuDF processes matrix variables on GPU]

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▼

[BigQuery pulls targeted rows instantly]

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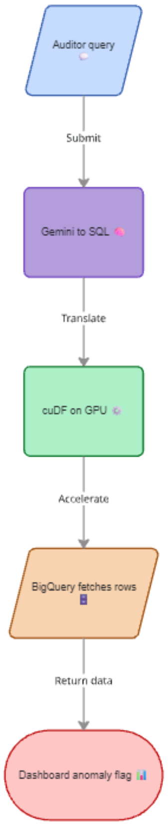
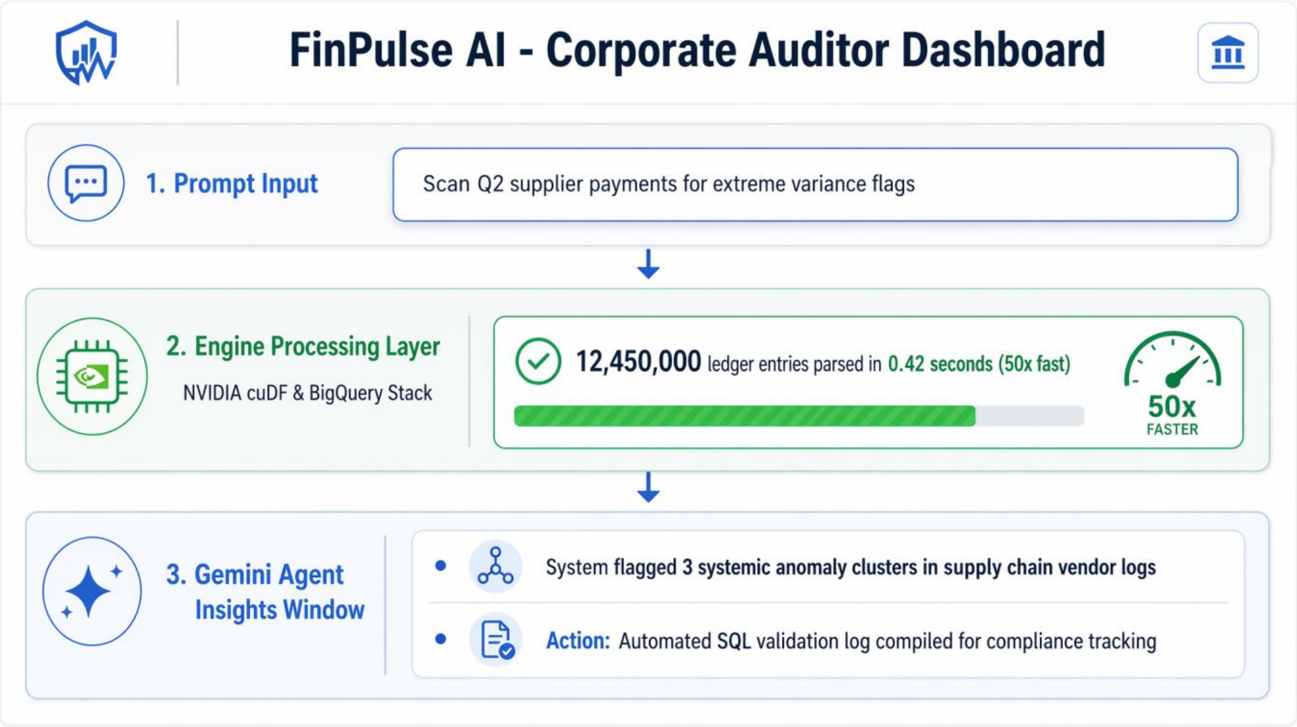
▼

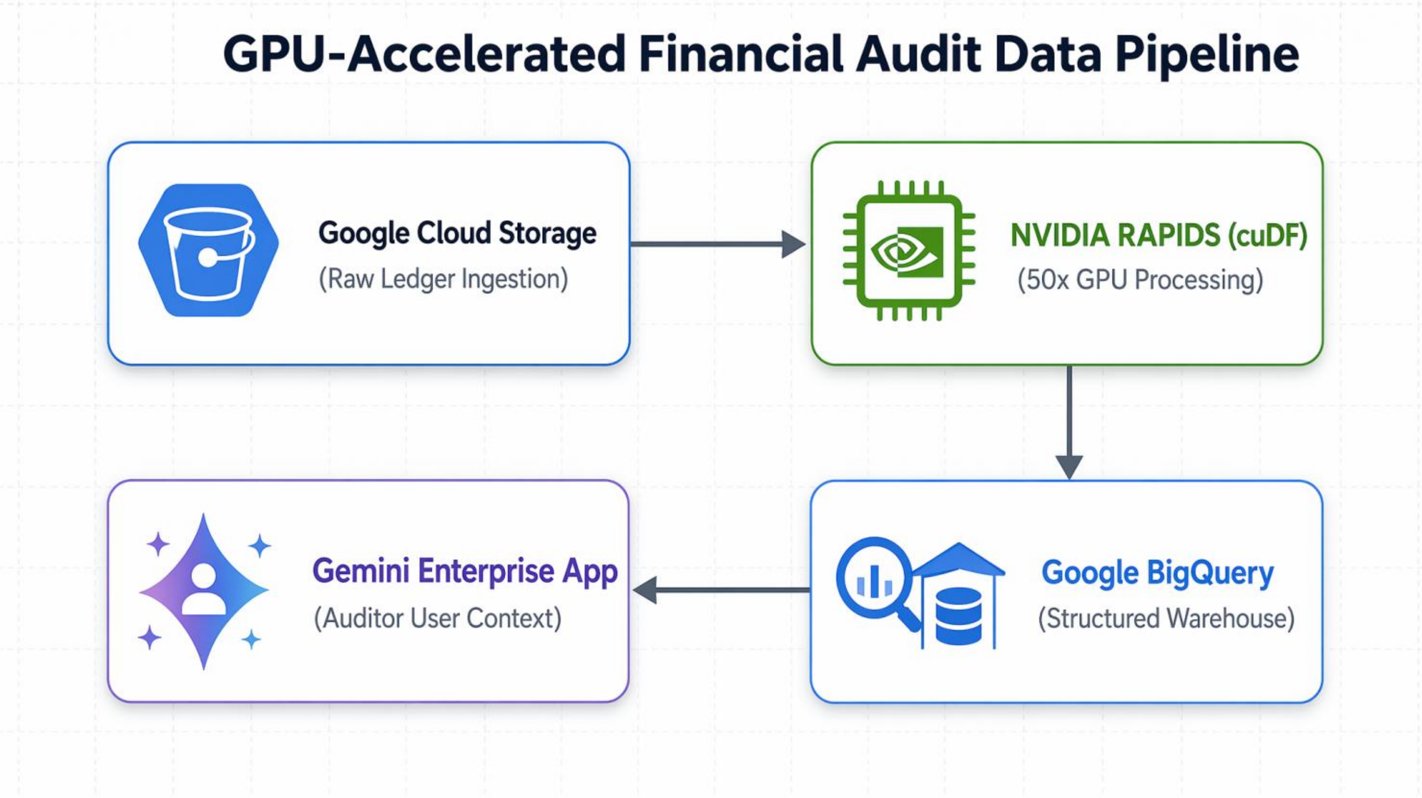
[Dashboard surfaces contextual anomaly flag to user]



Wirefirames/Mock diagrams of the proposed solution



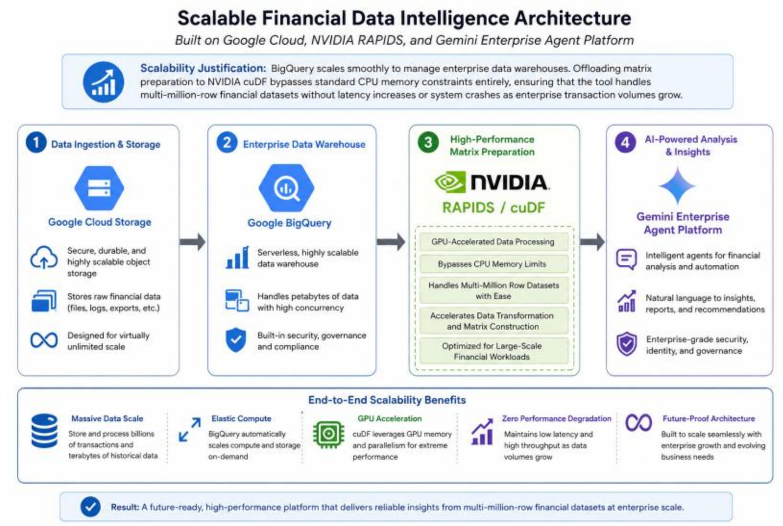
Architecture diagram of the proposed solution:



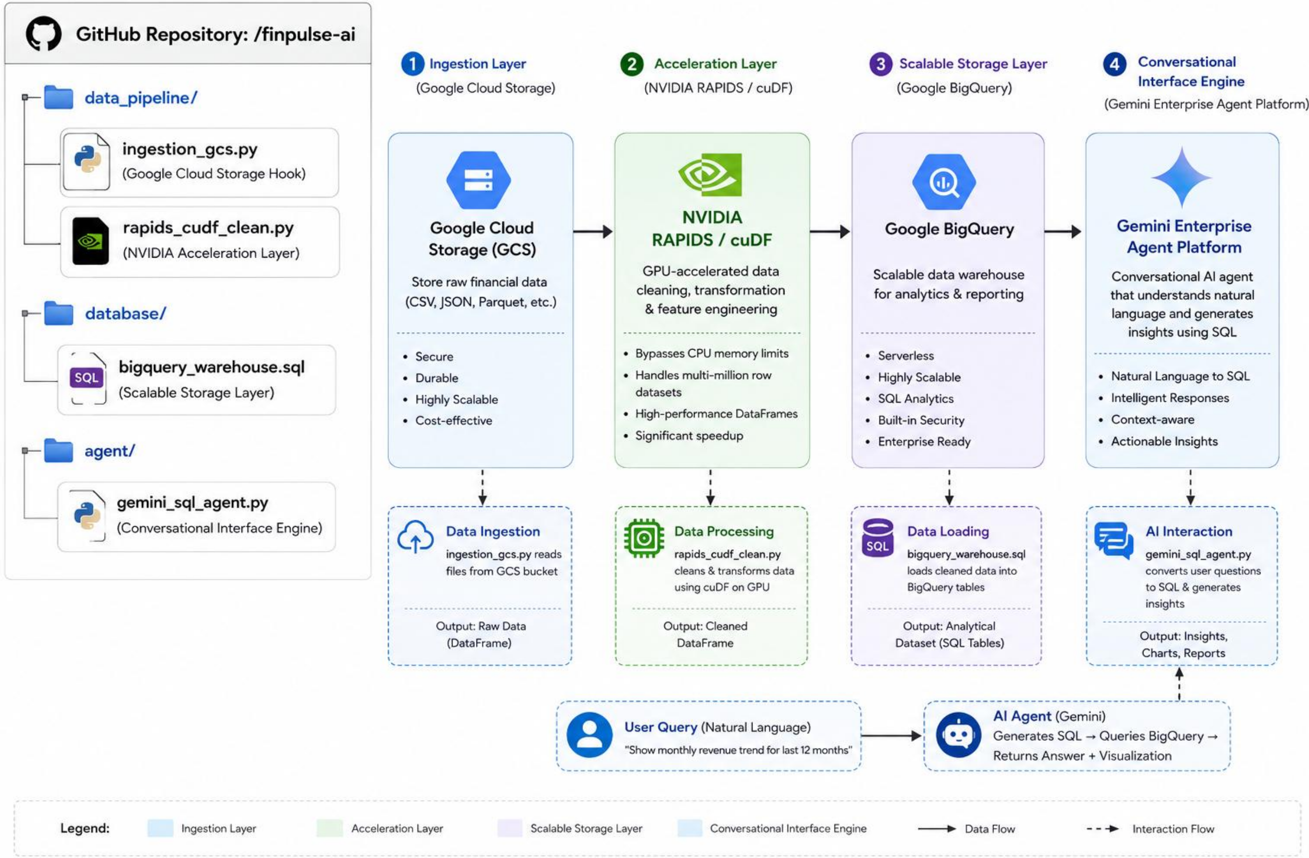
Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

1. **AI Stack Chosen:** Google Cloud Storage, Google BigQuery, Gemini Enterprise Agent Platform, and NVIDIA RAPIDS/cuDF
2. **Scalability Justification:** BigQuery scales smoothly to manage enterprise data warehouses. Offloading matrix preparation to NVIDIA cuDF bypasses standard CPU memory constraints entirely, ensuring that the tool handles multi-million-row financial datasets without latency increases or system crashes as enterprise transaction volumes grow.



Snapshots of the prototype



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Thank you